Position, distance, displacement

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Distinguish a location, a total path length, and a change in position.

SUCCESS CRITERIA - I can...

1. Label the origin, start, turn and finish.
2. Calculate path distance and signed displacement separately.
3. Explain why position and displacement can differ.

Core: teacher selects from 1-9. **Stretch:** 10 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “position” and “distance” to compare your model or result. Sentence starters are optional.

position: location measured from a chosen origin

distance: total length of the path traveled

displacement: change from initial position to final position, with direction

01 Warm-up: bring back an earlier idea

Positions are 15, 10, 5 m at 0, 2, 4 s.
Predict position at 7 s.
Give the rate, an assumption and a future test.

Sentence starter: I predict … . My model supports this because … .

Marzano check - circle: ① ② ③

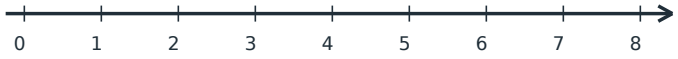
02 Worked example: explain the model

Reference: $\Delta x = x_f - x_i$; distance = sum of path lengths. Choose an origin and positive direction. i = initial; f = final.

Worked reasoning: A walker starts at 2 m, reaches 6 m, then ends at 4 m. Distance=4+2=6 m. Displacement=4-2=2 m right.

Your turn: The walker goes from the 2 m floor mark to the 6 m mark, then back to 4 m. On the number line, draw the two legs above the axis and a separate start-to-finish arrow below it.
Label each length.

Sentence starter: The … in my model represents … . Its label shows … .



Position (m); right is positive

Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

Position, distance, displacement

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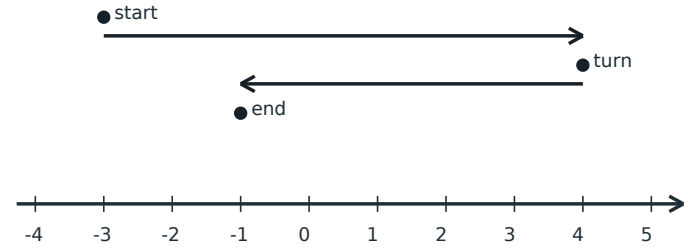
Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Track a signed route

A robot moves along an east-west test track. East is positive. It starts at -3 m, reaches +4 m, then stops at -1 m.
Use the route to find total distance and displacement.

Sentence starter: I found ... [unit]. This value means ... in this situation.



Position (m); right is positive

Marzano check - circle: 1 2 3

04 Change the coordinate convention

For the route in 3, reverse the positive direction while keeping the origin fixed.
Give the new endpoint coordinates and displacement. What stays unchanged?

Sentence starter: I notice ... . This supports ... because ... .

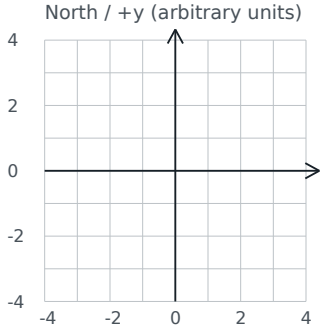
Marzano check - circle: 1 2 3

05 A closed path

A runner completes one 400 m lap of a circular track, then runs another 100 m.
Sketch the full trip and mark the final point. Is the displacement magnitude 500 m?
Explain how the track geometry determines the start-to-finish distance.

Word help: magnitude = size of a quantity without its direction

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .



East / +x (arbitrary units); each square = 1 arbitrary units

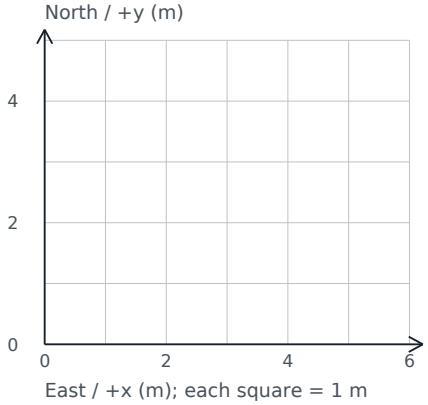
Marzano check - circle: 1 2 3

06 Different paths, same endpoints

A robot must travel from (0, 0) m to (4, 3) m.
Draw two routes on the grid: one using only east/north legs and one direct route.
Find each path length and the displacement magnitude both routes share.

Word help: magnitude = size of a quantity without its direction

Sentence starter: I found ... [unit]. This value means ... in this situation.



East / +x (m); each square = 1 m

Marzano check - circle: 1 2 3

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Continue down each column.

Sentence starters are optional. You may plan in any language.

07 Zero and nonzero

Can displacement be zero while distance is nonzero? Can distance be zero while displacement is nonzero? Justify both answers.

Word help: justify = support your answer with a reason or evidence

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: ① ② ③

08 Review: connect an earlier idea

A fictional observing report records 3 images of a moving object but gives no distance or size calibration.
State a supported observation and explain why a precise impact prediction is unsupported.
Separate what was observed from what can be inferred.

Word help: prediction = what a model says may happen next

Sentence starter: I predict ... . My model supports this because ... .

Observation	Supported claim	Missing information
____	____	____
____	____	____

Marzano check - circle: ① ② ③

09 Exit check

Start at +5 m, walk to -2 m, and finish at +1 m.
Calculate distance and displacement.
Explain why neither equals final position.

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: ① ② ③

10 STRETCH: test the model

In one dimension, start at -2 m and end at +3 m with total distance 11 m.
Find two possible single-turn positions and draw both routes.

Sentence starter: I found ... [unit]. This value means ... in this situation.



Position (m); right is positive

Marzano check - circle: ① ② ③